Independent Strategy Validation

Backtest, out-of-sample and overfitting review

Prepared for	Demo Client
Engagement	DEMO-0000
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Run ID	sv_20261111T175805Z_a12ef8

Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. No real securities or client data are used.

ProQuant Analytics was engaged by Demo Client to independently validate the **Dual moving-average trend (long-only)** strategy (fast=20,slow=100). We re-built the backtest from the client's data using our own engine and cost model, then tested whether the reported performance is likely to persist out of sample.

Overall assessment: **Validated with reservations - 2 point(s) to address**

Assessment against the thresholds in the validation scorecard below, agreed at kickoff. It is an opinion on the robustness of the backtest evidence, not a forecast or an investment recommendation.

NET CAGR	NET SHARPE	MAX DRAWDOWN
3.8%	**0.46**	**-28.9%**
gross 4.0%	benchmark 0.43	net of costs

DEFLATED SHARPE	PROB. OF OVERFITTING	WALK-FORWARD OOS SHARPE
67.6%	**2.8%**	**1.24**
45 trials; 95% required	CSCV	re-optimised yearly

Key findings

1. Net of modelled costs, the strategy returned 3.8% a year with a Sharpe ratio of 0.46 and a maximum drawdown of -28.9% over 2013-01-01 to 2025-12-31 versus 5.7% for the benchmark (Sharpe 0.43).
2. Costs reduce annual return by 0.22% at an annual turnover of 4.4x. The break-even cost is 97.4 bps per trade versus 5.0 bps assumed; doubling costs gives a Sharpe of 0.44.
3. Walk-forward re-optimisation over 9 annual folds produced an out-of-sample Sharpe of 1.24 (efficiency 1.53x); 4 different parameter sets were selected across folds.
4. In the hold-out period since 2022-01-01 the Sharpe ratio was 1.06 against 0.16 in the development period.
5. Accounting for 45 strategy variants tested, the Deflated Sharpe Ratio is 67.6% (95% required for significance); the expected best Sharpe from 45 zero-skill trials is 0.34.
6. The Probability of Backtest Overfitting is 2.8% across 12,870 combinatorial splits.
7. The look-ahead truncation test passed: removing future data does not change historical positions. Sharpe with one extra day of execution delay is 0.98x the base case.
8. Survivorship: Survivors-only CAGR 4.56% vs point-in-time universe 3.76%; 5 delisted names included.
9. Block-bootstrap Monte Carlo gives a 90% range for the Sharpe ratio of 0.02 to 0.93; the probability of a negative Sharpe is 4.5%. The 5th-percentile maximum drawdown is -37.1%.

Validation scorecard

Area	Test	Measured	Pass if	Result
Data	Data-quality errors remaining after cleaning	0	<= 0	PASS
Bias	Look-ahead: truncation test (weights unchanged when future data removed)	passed	>= 1	PASS
Bias	Sharpe with +1 day execution delay / base (lag 1)	0.98x	>= 0.7	PASS
Bias	Survivorship: CAGR overstatement of a survivors-only universe Survivors-only CAGR 4.56% vs point-in-time universe 3.76%; 5 delisted names included.	0.8%	<= 0.01	PASS
Performance	Net Sharpe ratio (after costs)	0.46	>= 0.5	WARN
Costs	Net Sharpe with 2x transaction costs	0.44	>= 0.25	PASS
Overfitting	Deflated Sharpe Ratio (45 trials)	67.6%	>= 0.95	WARN
Overfitting	Probability of Backtest Overfitting (CSCV)	2.8%	<= 0.2	PASS
Out-of-sample	Walk-forward efficiency (OOS / IS Sharpe)	1.53x	>= 0.5	PASS
Out-of-sample	Hold-out Sharpe / in-sample Sharpe	6.62x	>= 0.5	PASS
Robustness	Share of parameter grid with positive net Sharpe	100.0%	>= 0.8	PASS
Robustness	Worst neighbouring parameter Sharpe / base Sharpe	0.84x	>= 0.6	PASS
Robustness	Monte Carlo probability of negative Sharpe	4.5%	<= 0.05	PASS

Source: tables/scorecard.csv. WARN = between warning and pass thresholds; N/A = not applicable to this engagement.

Recommendations

- Net risk-adjusted return is modest; consider whether the strategy's role (diversifier, crisis hedge) justifies it, and size it accordingly.
- After adjusting for the number of variants tested, the Sharpe ratio is not significant at 95%. Extend the live/paper track record or reduce the research search space before scaling.

Methodology

This report sets out the verdict and the evidence behind it. The steps, formulas and references are in the methodology document. One thing helps when reading on: all headline results are after trading costs.

Results

1. Performance

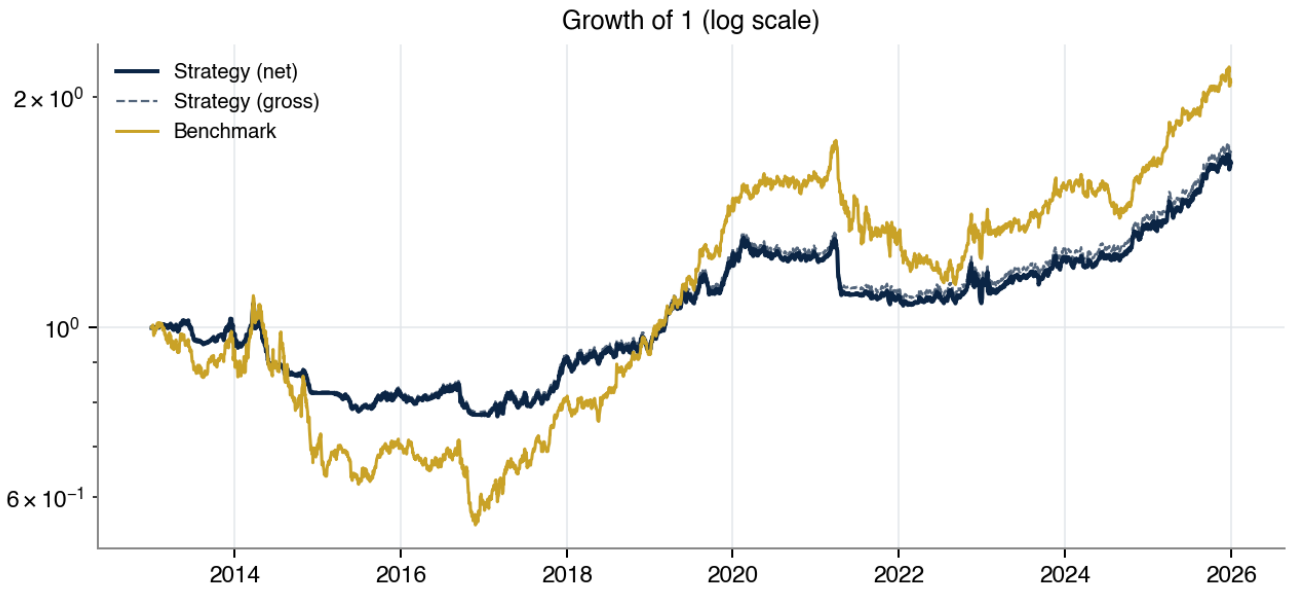


Figure 1. Growth of 1, net and gross of costs, against the benchmark. Source: tables/equity_curves.csv

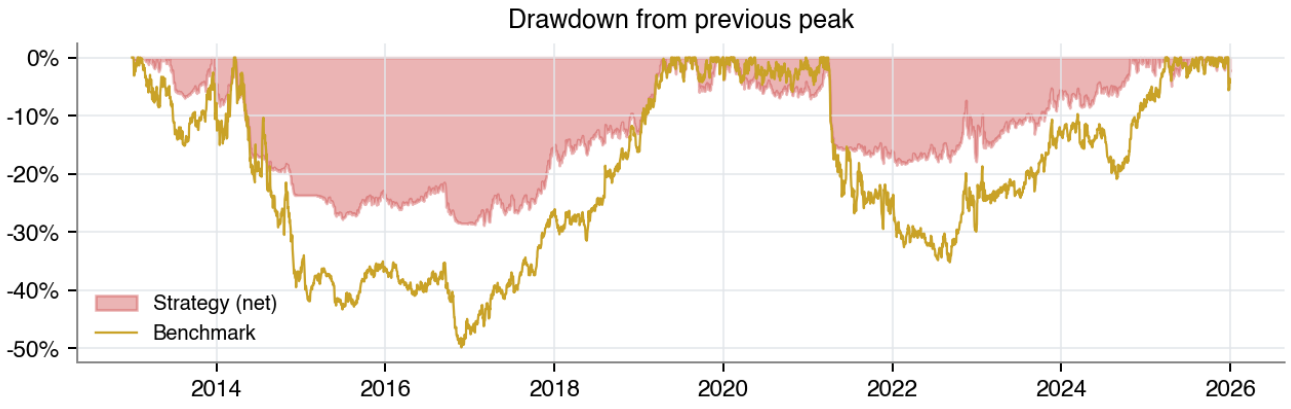


Figure 2. Drawdowns. Source: tables/drawdowns.csv

Metric	Strategy (net)	Strategy (gross)	Benchmark
CAGR	3.8%	4.0%	5.7%
Volatility	8.9%	8.9%	15.7%
Sharpe	0.46	0.49	0.43
Sortino	0.63	0.67	0.60
Max drawdown	-28.9%	-28.5%	-49.9%
Max drawdown duration (days)	1321	1314	1346
Calmar	0.13	0.14	0.11
Hit rate (days)	52.1%	52.3%	52.1%
Skew	-0.84	-0.84	-0.36
Excess kurtosis	10.23	10.23	4.95
VaR 95% (1-day)	0.8%	0.8%	1.4%
Expected shortfall 95% (1-day)	1.4%	1.4%	2.4%
Beta	0.45	0.45	n/a
Alpha (annual)	1.0%	1.3%	n/a
Information ratio	-0.26	-0.24	n/a
Annual turnover	4.43	4.43	n/a

Source: tables/metrics_summary.csv

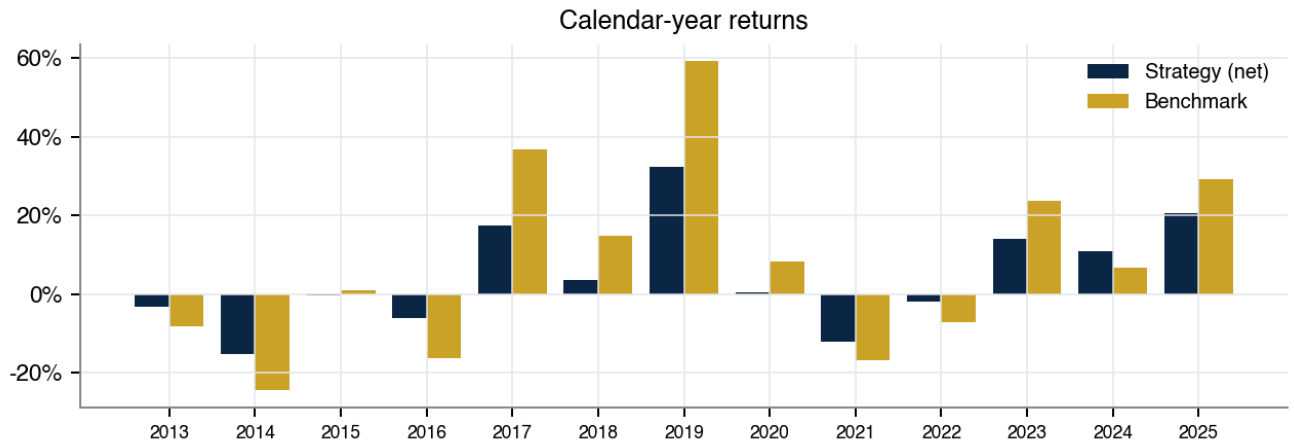


Figure 3. Calendar-year returns. Source: tables/daily_returns.csv



Figure 4. Rolling one-year Sharpe ratio. Source: tables/rolling_sharpe_1y.csv

Monthly net returns (%)

year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2013	-0.0	0.6	-0.5	-0.8	2.4	-3.7	-1.7	-0.5	1.7	0.5	1.3	-2.1	-3.0
2014	-2.3	2.5	4.5	-1.0	-10.4	-0.8	-0.6	-1.9	-0.3	1.2	-5.7	-0.8	-15.3
2015	0.0	0.0	-0.3	-0.2	-3.5	-1.4	1.7	-0.6	3.1	0.3	-0.9	1.9	-0.2
2016	-1.8	-0.8	1.3	-0.1	0.9	1.5	0.6	0.8	-5.2	-2.4	-0.8	0.1	-6.0
2017	0.5	0.3	2.3	2.8	-3.7	4.6	-3.6	2.7	-0.0	1.9	5.1	3.8	17.4
2018	-2.5	4.9	-1.3	1.6	-1.2	2.4	-2.3	3.1	-0.5	0.1	3.3	-3.6	3.6
2019	4.0	2.7	4.7	1.2	3.7	0.3	0.6	4.9	-4.2	-0.5	5.5	5.9	32.5
2020	1.0	2.0	-3.0	1.6	-1.0	-0.3	-1.0	-0.9	0.6	0.4	-0.1	1.2	0.5
2021	-1.5	1.8	1.7	-12.5	-0.5	0.4	-0.3	-1.2	-0.0	1.2	0.4	-1.2	-12.0
2022	-1.6	-0.6	1.9	-0.2	-0.7	2.2	-1.7	-0.2	1.4	3.5	2.7	-7.9	-1.8
2023	2.9	1.1	0.9	1.9	1.2	-0.1	0.7	0.1	0.3	1.8	1.2	1.4	14.2
2024	-1.5	1.1	-2.0	0.5	2.4	0.3	0.0	-0.8	1.7	6.9	-0.9	3.0	11.0
2025	-1.2	0.6	4.8	0.2	2.0	-0.0	2.9	4.4	3.6	-0.7	4.1	-1.6	20.6

Source: tables/monthly_returns_net.csv (net, %)

2. Transaction costs

Break-even cost: **97.4 bps** per trade, against 5.0 bps assumed (19.5x margin of safety).

Cost multiple	Per-trade (bps)	Net Sharpe	Net CAGR	Annual cost drag
0.0	0.0	0.49	4.0%	0.00%
0.5	2.5	0.47	3.9%	0.11%
1.0	5.0	0.46	3.8%	0.22%
2.0	10.0	0.44	3.5%	0.44%
3.0	15.0	0.41	3.3%	0.66%

Source: tables/cost_stress.csv

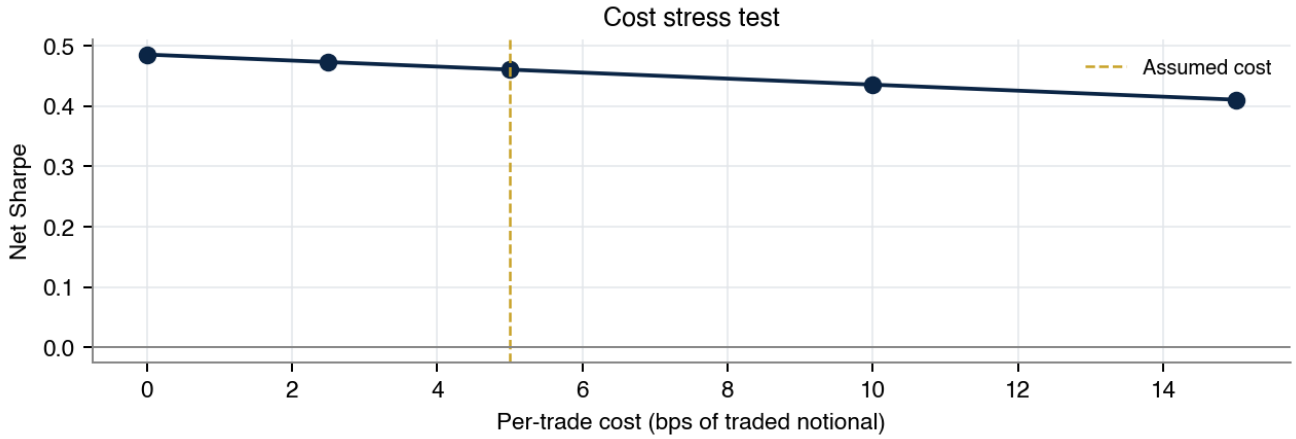


Figure 5. Net Sharpe as costs increase. Source: tables/cost_stress.csv

3. Out-of-sample: walk-forward and hold-out

Fold	Train from	Train to	Test from	Test to	Selected	IS Sharpe	OOS Sharpe
1	2013-01-01	2016-12-30	2017-01-02	2017-12-29	fast=5,slow=150	-0.35	1.90
2	2014-01-01	2017-12-29	2018-01-01	2018-12-31	fast=60,slow=250	0.35	0.46
3	2015-01-01	2018-12-31	2019-01-01	2019-12-31	fast=40,slow=250	0.80	3.30
4	2016-01-01	2019-12-31	2020-01-01	2020-12-31	fast=40,slow=250	1.61	0.48
5	2017-01-02	2020-12-31	2021-01-01	2021-12-31	fast=60,slow=250	1.70	-1.07
6	2018-01-01	2021-12-31	2022-01-03	2022-12-30	fast=60,slow=250	1.01	0.09
7	2019-01-01	2022-12-30	2023-01-02	2023-12-29	fast=60,slow=250	0.95	1.76
8	2020-01-01	2023-12-29	2024-01-01	2024-12-31	fast=60,slow=250	0.55	0.96
9	2021-01-01	2024-12-31	2025-01-01	2025-12-31	fast=60,slow=200	0.66	2.07

Source: tables/walk_forward_folds.csv

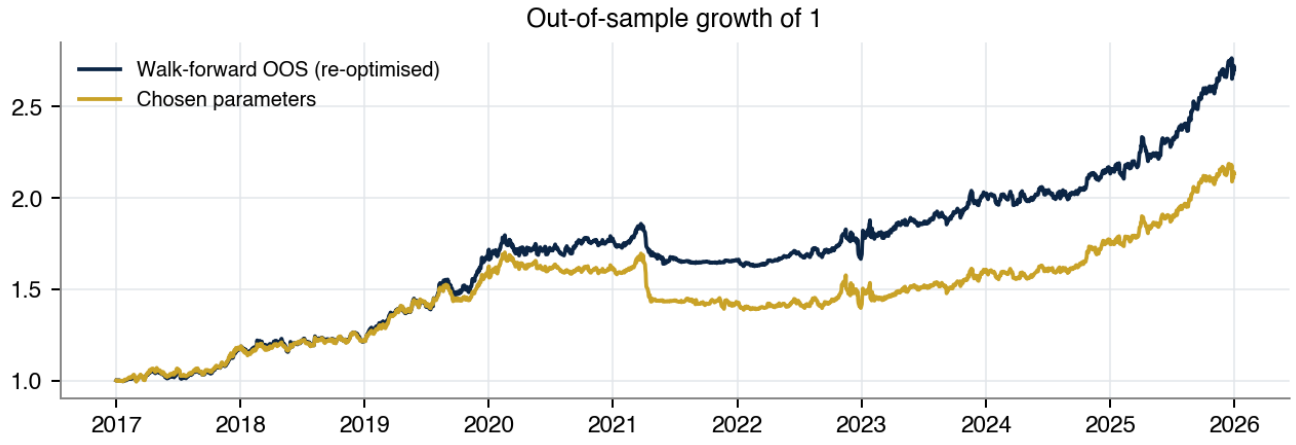


Figure 6. Stitched out-of-sample record. Source: tables/walk_forward_oos_returns.csv

Period	Start	End	Net Sharpe	Net CAGR	Max DD
in_sample	2013-01-01	2021-12-31	0.16	1.0%	-28.9%
holdout	2022-01-03	2025-12-31	1.06	10.3%	-11.4%

Source: tables/holdout.csv

4. Overfitting and bias

Look-ahead

Cut-off date	Max |weight diff|	Passed
2015-12-07	0.0e+00	True
2018-05-01	0.0e+00	True
2018-09-24	0.0e+00	True
2020-11-09	0.0e+00	True
2022-03-04	0.0e+00	True
2024-05-29	0.0e+00	True

Source: tables/lookahead_truncation_test.csv

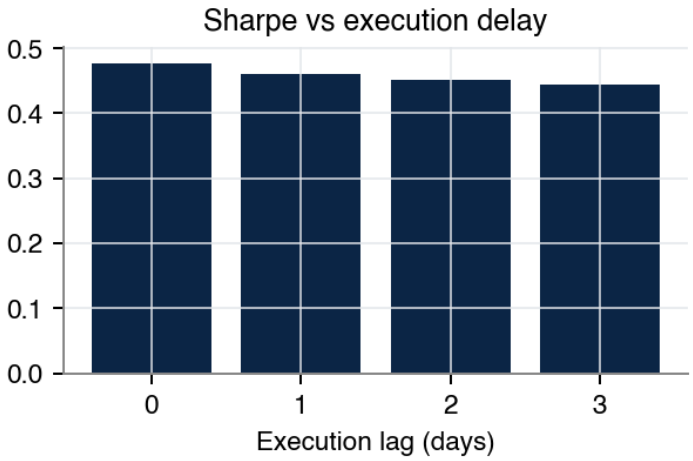


Figure 7. Net Sharpe by execution delay. Source: tables/timing_sensitivity.csv

Survivorship

Survivors-only CAGR 4.56% vs point-in-time universe 3.76%; 5 delisted names included.

Multiple testing

Annualised Sharpe (net)	0.46
Skew / kurtosis of daily returns	-0.84 / 13.23
Probabilistic Sharpe Ratio vs 0	95.2%
Trials (45), dispersion of trial Sharpes	0.15
Expected max Sharpe of zero-skill trials	0.34
Deflated Sharpe Ratio	**67.6%**
Minimum track record for 95% confidence (years)	13.1

Source: tables/sharpe_inference.csv

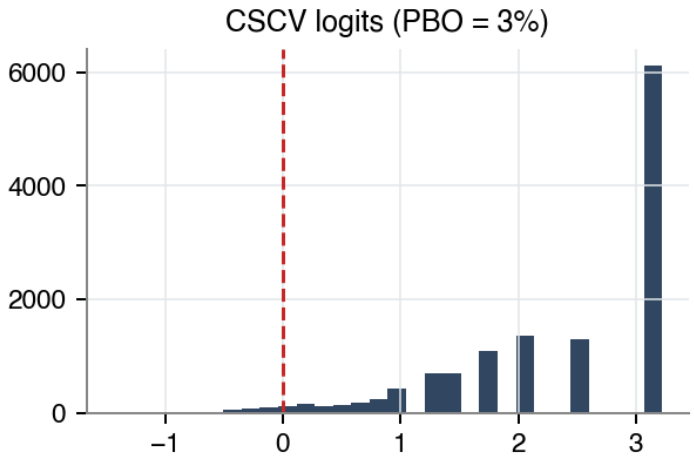


Figure 8. CSCV logit distribution; mass left of 0 = PBO. Source: tables/pbo_logits.csv

5. Parameter sensitivity

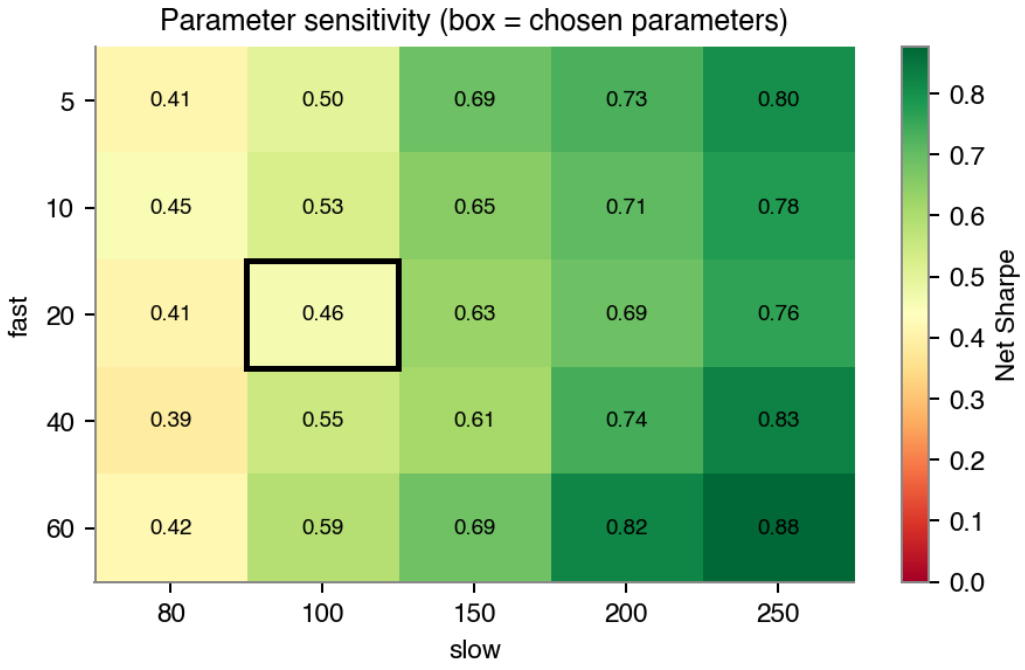


Figure 9. Net Sharpe across the parameter grid. Source: tables/sensitivity_grid.csv

Chosen parameters: Sharpe 0.46 (grid median 0.65, best 0.88); 100% of the grid has a positive net Sharpe; the weakest neighbouring setting retains 0.84x of the chosen Sharpe.

6. Monte Carlo

Metric	Observed	5th pct	Median	95th pct	P(< 0)
cagr	0.04	-0.00	0.04	0.08	5.8%
sharpe	0.46	0.02	0.46	0.93	4.5%
max_drawdown	-0.29	-0.37	-0.21	-0.13	n/a

Source: tables/monte_carlo_summary.csv

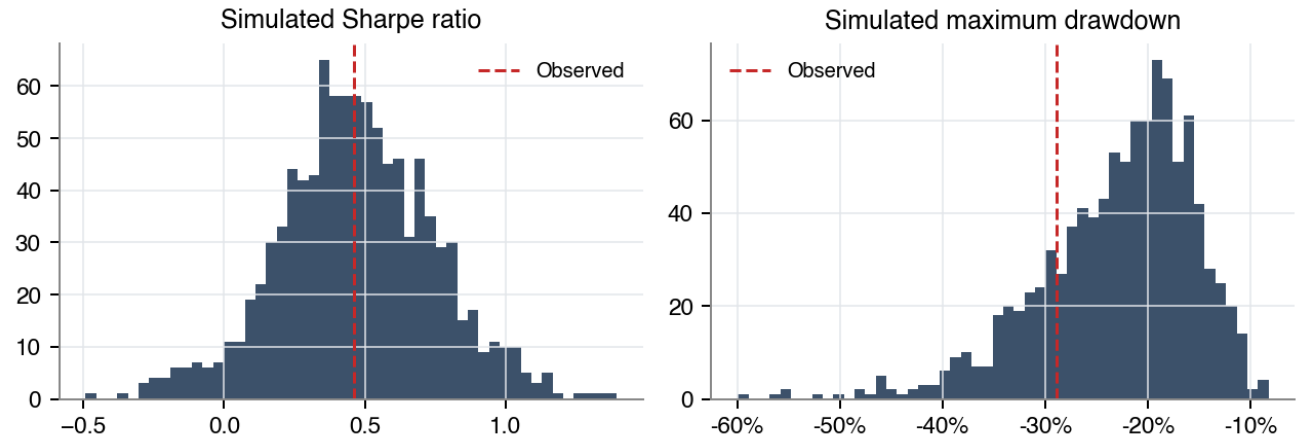


Figure 10. Bootstrap distributions with observed values. Source: tables/monte_carlo_simulations.csv

Limitations and assumptions

- Results depend on the data supplied by the client; we checked it for internal consistency but did not reconcile it to an independent vendor unless stated.
- The cost model is linear in traded notional. It does not model market impact that grows with order size, so results do not establish capacity. A capacity study requires volume data and the intended AUM.
- Daily close-to-close simulation: intraday execution, partial fills, halts and corporate-action timing are not modelled.
- Delisting returns are captured only to the last traded price; actual recovery values may differ.
- The Deflated Sharpe Ratio relies on the number of trials disclosed by the client; undisclosed research increases the true number of trials and lowers the true DSR.
- Bootstrap simulations resample the past; they do not represent regimes absent from the sample.
- Scorecard thresholds are conventions agreed at kickoff, not regulatory standards.
- This demonstration uses synthetic data; its numbers have no meaning for any real strategy.

Appendix

A.1 Configuration used

Strategy mode	reference_ma
Parameters	fast=20,slow=100
Execution lag	1 day(s)
Parameter grid	{'fast': [5, 10, 20, 40, 60], 'slow': [80, 100, 150, 200, 250]}
Hold-out start	2022-01-01
Walk-forward	{'train_years': 4, 'test_years': 1, 'anchored': False}
Monte Carlo	{'n_sims': 1000, 'mean_block_days': 20, 'seed': 42}

Source: config_used.yaml in the run folder

A.2 Data cleaning log (summary)

step	action	rows_affected	detail
duplicates	drop exact duplicate rows	2	
parse	missing close values in raw panel	6	
bad_ticks	set to missing (spike and reversal)	1	return +888.0% then -89.9%
stale_prices	flagged (kept) - identical close run	6	6 days unchanged
gaps	forward-filled interior gaps (max 3 days)	7	

Source: tables/cleaning_log.csv

Data quality

benchmark · 3653 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

prices · 99769 rows · PASSED (0 errors, 4 warnings)

Check	Severity	Column	Count	Message
missing_values	warning	close	6	6 missing (0.01%)
duplicate_rows	warning		2	2 fully duplicated rows
duplicate_keys	warning		4	4 rows share a key ['date', 'ticker'] (0 with conflicting values)
outliers	warning	close (daily return)	9	9 daily returns with robust |z| > 8.0

Source: data_quality/dq_prices.json

prices_clean · 99767 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

universe · 30 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	delisted_on	25	25 missing (83.33%)
missing_values	info	delisting_reason	25	25 missing (83.33%)

Source: data_quality/dq_universe.json

Audit trail

Run ID	`sv_20261111T175805Z_a12ef8`
Run started / finished (UTC)	2026-11-11T17:58:05+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`2703b230a84f (uncommitted changes) · core 0.1.0`
Config SHA-256	`dbf24edae564ce4e`
Input: prices	`99769 rows · sha256 b774dec28e812fb9`
Input: universe	`30 rows · sha256 d44d49659ae7dd4c`
Input: benchmark	`3653 rows · sha256 50d69ecafe672230`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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